

Lab: Testing Liveness Probe with NGINX Deployment

Lab Objective

By the end of this lab, you will be able to:

- Understand what a **Liveness Probe** is
 - Configure a **Liveness Probe** in an NGINX Deployment
 - Simulate application failure
 - Observe **automatic container restarts** triggered by Kubernetes
-

Concept Recap (Before Lab)

What is a Liveness Probe?

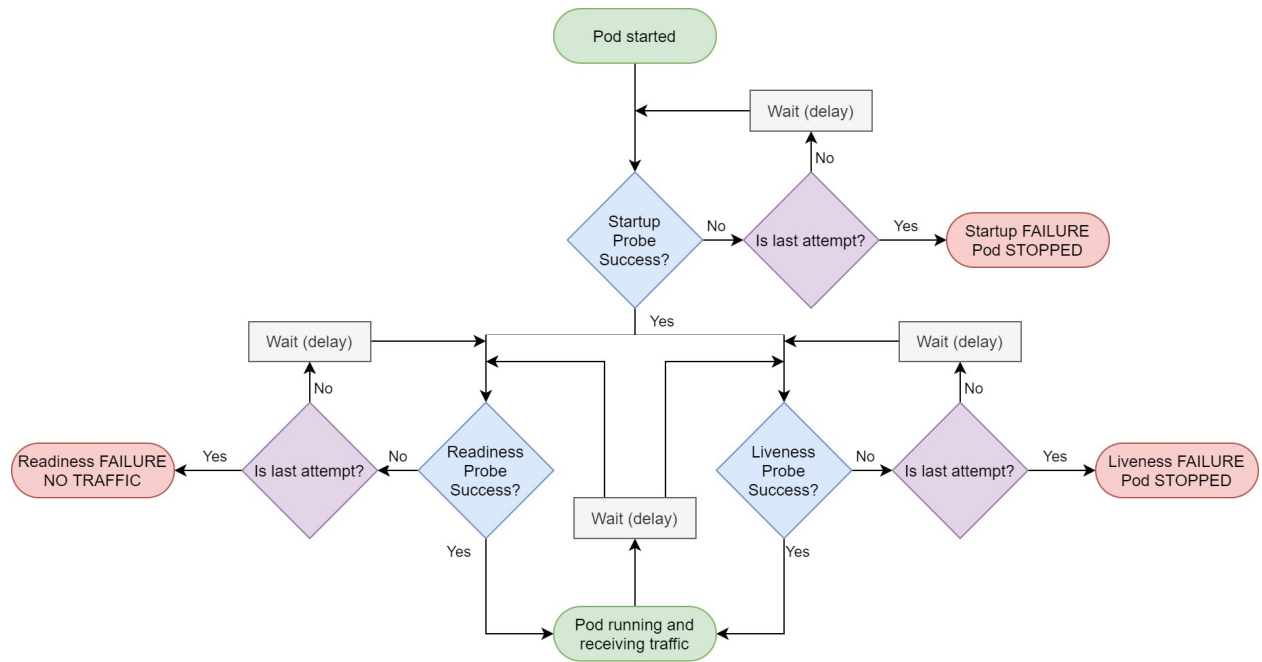
A **Liveness Probe** checks whether an application inside a container is **still running correctly**.

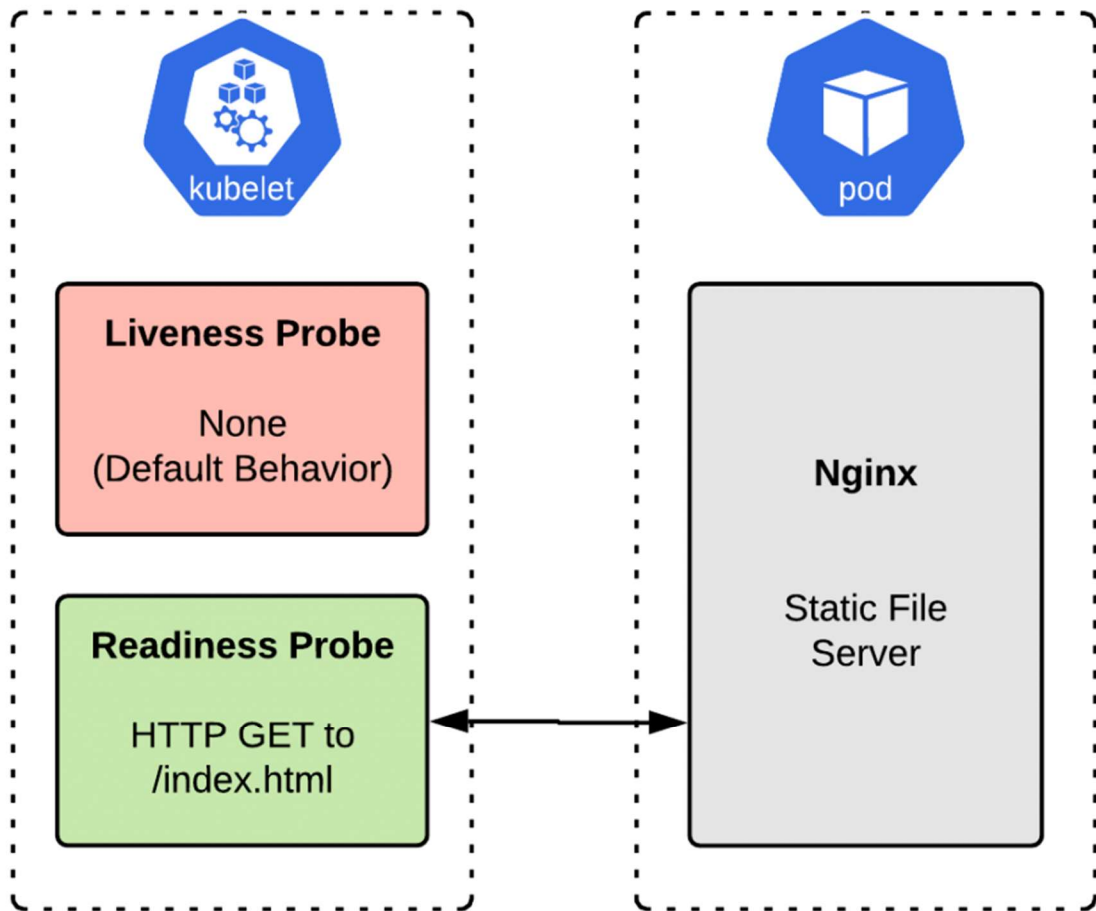
-  If probe **passes** → container continues running
-  If probe **fails repeatedly** → Kubernetes **restarts the container**

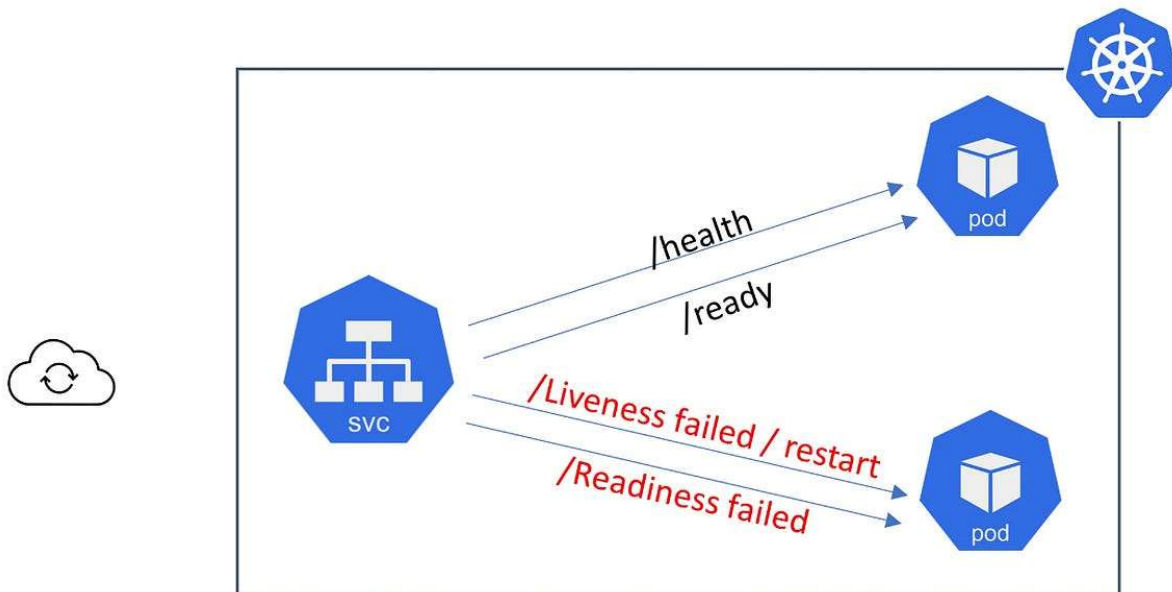
Important

Liveness probes are about **recovery**, not traffic routing.

How Liveness Probe Works (Flow)







Flow Explanation

1. Kubelet sends HTTP request to container
2. NGINX responds with HTTP 200 → Healthy
3. If NGINX stops responding → Probe fails
4. After failure threshold → Container is restarted

Lab Environment

Component	Value
Kubernetes Cluster	Minikube / Any K8s
Application	NGINX
Probe Type	HTTP GET
Port	80

Step 1: Create NGINX Deployment (Without Probe)

Create a file named **nginx-deploy.yaml**

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: nginx-liveness
```

```
spec:
```

```
  replicas: 1
```

```
  selector:
```

```
    matchLabels:
```

```
      app: nginx
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: nginx
```

```
    spec:
```

```
      containers:
```

```
        - name: nginx
```

```
          image: nginx:latest
```

```
          ports:
```

```
            - containerPort: 80
```

Apply Deployment

```
kubectl apply -f nginx-deploy.yaml
```

Verify Pod

```
kubectl get pods
```

Step 2: Add Liveness Probe to Deployment

Now update the deployment to include a **Liveness Probe**

Updated nginx-deploy.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: nginx-liveness

spec:

replicas: 1

selector:

matchLabels:

app: nginx

template:

metadata:

labels:

app: nginx

spec:

containers:

- name: nginx

image: nginx:latest

ports:

- containerPort: 80

livenessProbe:

httpGet:

path: /

port: 80

initialDelaySeconds: 10

periodSeconds: 5

failureThreshold: 3

Probe Parameter Explanation

Parameter	Meaning
httpGet	Check HTTP response
path: /	Root URL
initialDelaySeconds: 10	Wait 10s after start
periodSeconds: 5	Check every 5 seconds
failureThreshold: 3	Restart after 3 failures

Apply Updated Deployment

```
kubectl apply -f nginx-deploy.yaml
```

Step 3: Verify Liveness Probe

```
kubectl describe pod <pod-name>
```

Look for:

Liveness: http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3

Step 4: Simulate Application Failure

Now we **intentionally break NGINX** to test the probe.

Access the Pod

```
kubectl exec -it <pod-name> -- /bin/bash
```

Stop NGINX Inside Container

```
nginx -s stop
```

or delete index file:

```
rm /usr/share/nginx/html/index.html
```

Exit the container:

```
exit
```

Step 5: Observe Pod Restart

Watch Pod Status

```
kubectl get pods -w
```

You should see:

```
RESTARTS 1
```

Confirm via Events

```
kubectl describe pod <pod-name>
```

Look for:

Liveness probe failed

Killing container nginx

 **This confirms liveness probe is working**

Step 6: Verify Container Restart Count

```
kubectl get pods
```

Output example:

NAME	READY	STATUS	RESTARTS
nginx-liveness-xxx	1/1	Running	1

Key Observations

Scenario	Result
App running	No restart

Scenario	Result
App stops responding	Probe fails
Probe fails 3 times	Container restarted
Pod remains same	Only container restarts

⚠ Common Mistakes (Exam + Interview)

- ✗ Using **liveness probe** for slow-starting apps
 - ✗ Very small `initialDelaySeconds`
 - ✗ Using liveness probe for dependency checks (DB, API)
 - ✓ Use **readiness probe** for traffic control
 - ✓ Use **liveness probe** only for deadlock detection
-

🧠 Interview Question Mapping

- Q: What happens if liveness probe fails?
A: Kubernetes restarts the container automatically.
- Q: Does liveness probe affect Service routing?
A: ✗ No (Readiness probe does)
-

✅ Lab Completion Checklist

- Deployment created
- Liveness probe added
- Failure simulated
- Container restarted automatically
- Events verified